NLPLarpers NewsBot Individual Journal

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Individual journal example to personalize before submission

GitHub repository: [Insert your public repository URL]

Project purpose and individual contribution

I worked individually on a NewsBot prototype using the BBC News Classification dataset. I downloaded the CSV directly from Kaggle and uploaded it to Google Colab. The file contained 1,490 articles in business, entertainment, politics, sport and tech. The goal was to combine text processing, classification, sentiment and entity extraction into one workflow that could help organize news and support media monitoring.

Technical learning and integration challenges

The most challenging part was adapting the starter notebook to the real dataset. Its sample data included titles, dates and sources, but the BBC CSV supplied only article IDs, text and categories. I renamed the columns and removed references to unavailable fields. This taught me to inspect the actual data instead of assuming that every template field exists.

Missing NLTK data also caused errors. I added the downloads below for tokenization and English POS tagging. I then reran the affected cells instead of rewriting unrelated functions.

```
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng')
```

Naive Bayes failed because the VADER compound feature can be negative. A MinMaxScaler inside its pipeline converted inputs to nonnegative values. A later prediction error revealed another integration issue: training used 5,000 TF-IDF features and four sentiment values, while the template supplied seven dummy values for new articles.

```
scores = self.analyze_sentiment(full_text)
sentiment_features = np.array([[scores['compound'],
   scores['pos'], scores['neu'], scores['neg']]])
features_combined = np.hstack([
   features.toarray(), sentiment_features])
```

I also updated the call to classify_article(processed_text, full_text). Training and prediction then both used 5,004 features in the same order. The final demonstration successfully processed all three template articles.

Results and what they mean

The saved comparison reported test accuracy of 96.64% for Naive Bayes, 97.32% for Logistic Regression and 95.64% for SVM on 298 articles. These are preliminary figures: TF-IDF was fitted before the split, and the same test scores selected the best model. A data-quality check also found 50 repeated article texts. A stronger evaluation must remove duplicates before splitting, fit features inside training folds and select settings without using final test results.

The test examples showed why accuracy alone is insufficient. The Microsoft acquisition story was classified as tech with 37.42% probability and could also fit business. The Lakers story was sport at 69.03%. The climate story was politics at only 25.97%; the model had no environment category. Its slightly positive VADER score also conflicted with the alarming wording. These examples were fictional template tests, not verified news reports.

Business value and ethical considerations

An editor could use the system for initial article routing, while an analyst could search for names, organizations, and money amounts. The saved NER analysis extracted 42,031 mentions. However, mentions are model predictions: the test output incorrectly grouped “Microsoft Acquires” as an organization. Human review is needed before connecting people or companies to events.

Entertainment had the highest mean VADER sentiment, 0.5991. Politics had the lowest, 0.0578, which was still slightly positive. These scores describe language in this dataset; they do not prove media bias or measure public opinion. Automated labels can mislead readers if uncertainty, dataset limitations and extraction errors are hidden.

Future improvements and professional development

My next priorities would be training-only feature fitting, independent full-length test articles, clearer handling of unfamiliar topics and better review of entity spans. I would compare TF-IDF alone with added sentiment features before claiming that the added features improve performance. Topic modeling and contextual language models are possible future extensions. The current sentiment component measures polarity rather than separate emotions such as fear or anger.

This project helped me practice reading tracebacks, checking column names, and matching features across a pipeline. For my portfolio, I would show the notebook, setup instructions, charts, measured limitations and examples of both useful and uncertain predictions. I would identify the course template and AI-assisted debugging, explain my changes in my own words, and include the actual public GitHub link. Group collaboration does not apply to this individual project.

Sources: supplied ITAI 2373 starter notebook and assignment; BBC News Classification on Kaggle, kaggle.com/competitions/learn-ai-bbc/data. Numerical results above come from the saved notebook outputs.